

Machine Learning — Main Topics

A structured roadmap from fundamentals to professional Machine Learning projects.

1. Python for Machine Learning

- Python basics, data types, operators, conditions, loops, functions
- Lists, tuples, dictionaries, sets, strings
- Functions, modules, exceptions, file handling
- Object-Oriented Programming and virtual environments

2. NumPy

- Arrays and dimensions
- Indexing and slicing
- Reshaping and broadcasting
- Vectorized operations and basic numerical statistics

3. Pandas

- Series and DataFrame
- Reading CSV/Excel data
- Filtering and sorting
- GroupBy and aggregation
- Merge, join and concatenate
- Missing values and duplicates

4. Data Cleaning & Preprocessing

- Missing-value treatment
- Duplicate handling
- Outlier detection
- Encoding categorical variables
- Feature scaling and normalization

5. Exploratory Data Analysis (EDA)

- Univariate, bivariate and multivariate analysis
- Distribution analysis
- Correlation
- Feature relationships and business insights

6. Data Visualization

- Matplotlib
- Seaborn
- Bar charts, histograms, boxplots, scatter plots
- Heatmaps and model-result visualization

7. Statistics for ML

- Mean, median, mode
- Variance and standard deviation
- Probability
- Distributions
- Correlation and covariance
- Sampling and hypothesis testing basics

8. Machine Learning Fundamentals

- Supervised vs Unsupervised Learning
- Features and target
- Training, validation and test sets
- Overfitting and underfitting
- Bias-variance trade-off
- Cross-validation

9. Supervised Learning

- Linear Regression
- Multiple Linear Regression
- Logistic Regression
- K-Nearest Neighbors
- Decision Trees
- Random Forest
- Support Vector Machines
- Naive Bayes
- Gradient Boosting

10. Unsupervised Learning

- K-Means Clustering
- Hierarchical Clustering

- DBSCAN
- Dimensionality reduction with PCA

11. Feature Engineering

- Feature selection
- Transformation
- Encoding
- Scaling
- Polynomial features
- Interaction features

12. Model Evaluation

- Regression: MAE, MSE, RMSE, R^2
- Classification: Accuracy, Precision, Recall, F1-score, ROC-AUC
- Confusion Matrix
- Cross-validation

13. Hyperparameter Tuning

- Grid Search
- Random Search
- Model pipelines
- Choosing parameters without data leakage

14. Deep Learning Basics

- Neurons and layers
- Activation functions
- Loss functions
- Backpropagation
- Optimizers
- ANN
- CNN and RNN fundamentals

15. ML Projects & Deployment

- Problem definition
- Data pipeline
- Model training
- Model serialization

- FastAPI/Streamlit deployment
- Git/GitHub and professional README
- Basic MLOps concepts

Recommended Learning Order

Python → NumPy → Pandas → Data Cleaning → EDA → Visualization → Statistics → ML Fundamentals → Supervised Learning → Unsupervised Learning → Feature Engineering → Evaluation → Tuning → Projects → Deployment → Deep Learning